

When Geometry Is Not Enough: Object-Dependent Failure Modes of Learning-to-Rank Grasp Selection in Open-World Manipulation

Abstract

Learned grasp reranking promises to improve open-world manipulation, yet practitioners lack principled guidance on when such methods reliably help or harm. We present LGGSN, a lightweight pairwise-trained MLP that reranks GR-ConvNet candidates using geometric and query-conditioned features within a language-conditioned pipeline. Training with Bayesian Personalised Ranking (BPR) loss and augmenting the base feature set with episode-relative context and object query embeddings improves offline ranking accuracy from 0.447 to 0.750. Through a systematic paired evaluation over 175 trials (25 seeds $\times$ 7 objects, MuJoCo/SO-ARM101), we establish a geometric predictability framework: objects with low flat fraction (`flat_frac` < 0.1) yield consistent reranking gains (PowerDrill +4, MustardBottle +2), while geometrically degenerate objects (`flat_frac` > 0.4) cause feature saturation that nullifies ranking signal. Spearman correlation between `flat_frac` and reranking outcome reaches $\rho = -0.96$ on PyBullet ($p < 0.01$) and $\rho \approx -0.5$ on MuJoCo, where thin objects exhibit an execution floor effect rather than a ranking failure, enabling deployment-time prediction of reranker utility without trial execution. Our work provides the first systematic characterization of geometry-dependent reranking behavior in open-world grasping, and establishes concrete requirements for successor methods targeting degenerate object categories.

1. Introduction

Language-conditioned robot manipulation requires not only identifying *which* object to grasp but also selecting a grasp pose that will physically

succeed. Vision-based candidate generators such as GR-ConvNet [1] produce a ranked list of candidate grasps, but their quality estimates are computed independently per image patch, without considering the full geometry of the episode or the semantic properties of the target object. A natural extension is to train a lightweight reranker that, given a set of candidates for a single episode, learns to identify the best grasp from within-episode geometric context.

Prior work on grasp quality estimation largely operates offline on fixed object sets with known geometry [2, 3]. In open-world settings, where object identity is specified at runtime via natural language, the reranker must generalise across a diverse object distribution without access to CAD models or object-specific supervision. This introduces a fundamental challenge: the features that discriminate good from bad grasps depend heavily on object geometry, yet the reranker must be trained on a shared feature space.

We design and evaluate LGGSN within the OWG (Open-World Grasping) pipeline [4], which uses a Vision-Language Model (VLM) grounder to localise objects and GR-ConvNet to generate grasp candidates. Our contributions are:

1. **A pairwise BPR training procedure** for grasp reranking from live robot data, which avoids the within-episode label degeneracy that causes BCE training to collapse.
2. **Two episode-relative context features** (`dist_to_centroid`, `z_rel`) that address feature collapse in top-down scenarios and improve offline pair accuracy by 49 %.
3. **A paired empirical evaluation** over 25 seeds $\times$ 7 objects (175 trials) that reveals the net task effect of reranking is near zero, driven

by object-dependent failure modes predictable from geometric statistics.

4. **A diagnostic framework** correlating per-object geometric properties (flat fraction, height spread, yaw spread) with reranking benefit, explaining both where the method helps and where it harms.

This paper reports a predominantly negative result: the system does not reliably improve overall task success. Its value lies in a reproducible geometric diagnosis of *why* geometry-only reranking fails for specific object categories, and in the concrete design requirements this diagnosis exposes for future systems.

2. Related Work

2.1. Grasp Quality Estimation

GraspNet-1Billion [3] and related methods learn grasp quality predictors from large annotated datasets of 3-D point clouds, using object-centric representations. Dex-Net [2] formulates grasp quality as a function of contact geometry and friction, training on simulated perturbation outcomes. AnyGrasp [11] extends this paradigm with a graspness representation that generalises across unseen objects in cluttered scenes, still assuming access to dense point-cloud geometry. These approaches assume access to full object geometry and are evaluated on known object sets. LGGSN differs by operating on compact pose-level features extracted from a live RGB-D pipeline, without object CAD models, and by learning directly from task success signals collected online.

2.2. Pairwise and Listwise Learning to Rank

Bayesian Personalised Ranking (BPR) [5] was originally proposed for collaborative filtering and has since been applied across recommendation and information retrieval. Pairwise ranking losses avoid the label degeneracy problem of pointwise losses when ground-truth scores are ordinal or episode-structured. In grasping, [6] applies listwise ranking to offline grasp datasets. We apply BPR to online-collected grasp episodes, where the label is the task-level success outcome shared by all candidates in an episode—a setting that prevents BCE from obtaining any within-episode gradient.

2.3. Learning from Robot Interaction

DexGraspNet 2.0 [7] and RoboAgent [8] demonstrate learning grasp policies directly from robot experience. Our setting is lower data (605 episodes, 7 objects) and targets a plug-in reranker rather than an end-to-end policy. The closest prior work in this regime is [9], which trains a grasp success predictor from live trials; we extend this direction to the pairwise setting and study the failure modes that emerge at the level of individual object categories. Concurrent work on learning-based grasp selection from interaction data includes [16, 17], which focus on task-level success in cluttered scenes but do not analyse per-object geometric failure modes.

2.4. Language-Conditioned and Open-World Grasping

Language-conditioned manipulation has advanced rapidly with large vision-language models (VLMs). CLIPort [10] uses CLIP features for semantic pick-and-place without object-specific models. RT-2 [12] and OpenVLA [13] co-fine-tune VLMs on robot action data, achieving broad language-following but at the cost of high compute and limited grasp-pose precision. GraspGPT [14] grounds natural-language task descriptions to grasp affordances, focusing on *which* grasp to choose rather than improving a candidate generator’s quality estimates. SpatialVLM [15] enhances spatial reasoning in VLMs for manipulation, but operates at the policy rather than the reranker level. OWG [4] and its Stage 4 reranker provide the pipeline we augment; LGGSN is the only component using learned geometric reasoning beyond the VLM grounder and GR-ConvNet generator. To our knowledge, no prior work provides a systematic failure analysis of learned reranking across diverse object geometries in an open-world setting.

3. Method

3.1. System Overview

The OWG pipeline has two stages relevant to this work:

- **Stage 3 (baseline):** VLM grounder localises the target object; GR-ConvNet generates N grasp candidates; the top-ranked candidate by GR-ConvNet score is executed.
- **Stage 4 (proposed):** Same candidate gener-

ation; LGGSN reranks the N candidates; the top-ranked candidate by LGGSN score is executed.

LGGSN is a lightweight MLP trained offline on live-collected episode data and loaded at inference time.

3.2. Feature Representation

Each grasp candidate is represented by an 18-dimensional feature vector.

Base features (12-dim): grasp position (x, y, z) ; orientation (ϕ, θ, ψ) in world-frame RPY; gripper opening **width**; GR-ConvNet quality **score**; pre-grasp vertical offset **dz**; post-grasp lift **dz_lift**; collision-avoidance flag **need_dz**; estimated object height at grasp point H .

Episode-relative context features (2-dim, v2):

- **dist_to_centroid:** L2 distance from the candidate’s (x, y) to the episode centroid $(\bar{x}, \bar{y})$. Encodes how central or peripheral the approach is.
- **z_rel:** min-max normalised height within the episode, $(z - z_{\min}) / (z_{\max} - z_{\min} + \varepsilon)$. Encodes relative vertical position within the episode’s height range.

Object query embedding (4-dim, v5d):

A learned embedding $\mathbf{e}_q \in \mathbb{R}^4$, implemented as `nn.Embedding(7, 4)`, is appended to the 14-dimensional geometric features and trained jointly with the MLP. The class index $q \in \{\text{banana, can, pear, mustard, drill, cracker, cylinder}\}$ is derived from the VLM grounding output. The combined input $\mathbf{x} = [\mathbf{x}_{\text{geom}}; \mathbf{e}_q] \in \mathbb{R}^{18}$ allows LGGSN to adapt feature weighting per object class. At inference, an unknown class uses the mean over all seven embeddings.

Motivation. In typical top-down grasping, five of the twelve base features— x , y , ϕ , θ , and **width**—are nearly constant across all candidates in a single episode (the gripper approaches from directly above at a fixed overhead position). This leaves only 7 effectively informative geometric dimensions for within-episode ranking; the query embedding adds 4 object-conditional dimensions. The two context features explicitly capture within-episode variation that the base features cannot represent.

3.3. LGGSN Architecture

LGGSN is a 2-layer MLP:

$$f(\mathbf{x}) = \mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + b_2 \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^{18}$, $\mathbf{W}_1 \in \mathbb{R}^{40 \times 18}$, $\mathbf{W}_2 \in \mathbb{R}^{1 \times 40}$. Total parameters: 829 (including $7 \times 4 = 28$ for the query embedding table). At inference time, $\sigma(f(\mathbf{x}))$ is used as the quality score.

3.4. BPR Training

Dataset construction. Episodes are collected by running the OWG pipeline in MuJoCo simulation with the SO-ARM101 manipulator. Each episode produces N grasp candidates for a given (object, scene) pair and receives a binary label based on whether the executed grasp succeeded. Grounding-failed episodes (VLM API timeout or HTTP error) are identified by cross-referencing the batch log and excluded from training, as their negative labels reflect network failures rather than geometric failures.

Pairwise label. For a pair of episodes (p, n) with the same query object where p succeeded and n failed, each candidate in p is treated as a *positive* and each candidate in n as a *negative*. This cross-episode pairwise signal is the only supervision available: within a single episode all candidates share the same label, so no within-episode gradient exists under a pointwise loss.

Loss. The BPR loss for a (positive, negative) candidate pair is:

$$\mathcal{L}_{\text{BPR}} = -\log \sigma(f(\mathbf{x}_+) - f(\mathbf{x}_-)) \quad (2)$$

Training details. Adam optimiser, $\text{lr} = 10^{-3}$, batch size 32, 30 epochs, 80/20 episode-level train/val split.

3.5. Geometry-Conditioned Gate (GC-LGGSN)

As a negative-result variant, we trained a GC-LGGSN in which a 3-dim episode context $\mathbf{z} = [\text{flat_frac}, \sigma_H, \sigma_\psi]$ gates the 14-dim feature vector via a learned sigmoid mask (+302 parameters, trained jointly under eq. (2)). Results are in section 4.7.

4. Experiments

4.1. Evaluation Protocol

All experiments use a *paired* evaluation: for each (seed, object) pair, Stage 3 and Stage 4 are run on the same MuJoCo/SO-ARM101 scene. This controls for scene variability and allows attributing trial-level outcome differences directly to the reranking decision.

Metric.

$$\text{net} = \underbrace{|\{i : S4_i=1 \wedge S3_i=0\}|}_{\text{impr.}} - \underbrace{|\{i : S3_i=1 \wedge S4_i=0\}|}_{\text{regr.}} \quad (3)$$

Scale. Primary evaluation: 25 seeds $\times$ 7 objects = 175 paired trials. Ablation studies: 10 seeds $\times$ 6 objects = 60 trials.

Objects. Banana, CrackerBox, MustardBottle, Pear, PowerDrill, Scissors, TomatoSoupCan—seven YCB objects spanning flat/thin, cylindrical, box-like, round, and asymmetric-tool geometries.

4.2. Training Data

The candidate log `lggsn_live_candidates.jsonl` contains 3,026 candidate records from live episode collection covering all seven objects. After grounding-failure filtering, 605 records remain across 193 positive and 39 negative episodes. Pairs are formed as the Cartesian product of all (positive episode, negative episode) pairs per query object. The positive-to-negative episode ratio is approximately 5:1 (193 vs. 39), meaning each negative episode participates in many more pairs than each positive; this imbalance is inherent to the online collection setting (successes are more common than failures once the pipeline is functional) and is not re-weighted during training.

4.3. Main Result

Table 1 reports the Stage 3 vs. Stage 4 comparison on MuJoCo/SO-ARM101. The deployed checkpoint is v5d (18-dim features with query embedding, val pair_acc 0.750; see table 3 row E). The per-object breakdown in table 1 uses v2 (14-dim) data to include all seven objects; v5d gains on the 5-object training subset are reported in table 3. The aggregate net improvement under v2 is +1: PowerDrill improves consistently (+4), while CrackerBox (−2), MustardBottle (−1), and Pear (−1) regress. Scissors registers zero success on both

Table 1. Stage 3 vs. Stage 4 on MuJoCo/SO-ARM101 — 25 seeds $\times$ 7 objects (175 paired trials). Bold indicates largest positive/negative net.

Object	S3	S4	Net
Banana	25/25	25/25	0
TomatoSoupCan	25/25	25/25	0
Pear	25/25	24/25	−1
MustardBottle	16/25	15/25	−1
Scissors	0/25	0/25	0
CrackerBox	10/25	8/25	−2
PowerDrill	16/25	21/25	+4
Total	117/175	118/175	+1

Net = S4 successes − S3 successes (paired).

Table 2. Cross-platform validation: Stage 3 vs. Stage 4 on PyBullet/Panda — 25 seeds $\times$ 6 objects (150 paired trials). Bold indicates largest positive/negative net. Pear is excluded because a calibrated PyBullet mesh model was not available at the time of cross-platform evaluation.

Object	S3	S4	Im.	Re.	Net
Banana	18/25	18/25	0	0	0
CrackerBox	13/25	17/25	4	0	+4
MustardBottle	21/25	23/25	2	0	+2
PowerDrill	22/25	21/25	0	1	−1
Scissors	20/25	16/25	0	4	−4
TomatoSoupCan	21/25	20/25	0	1	−1
Total	115/150	115/150	13	13	0

Im. = improvements; Re. = regressions.

stages, a floor effect that provides no reranking signal. Stage 3 achieves 66.8% (117/175) and Stage 4 achieves 67.4% (118/175). Figure 1 visualises the per-object breakdown. Cross-platform validation against PyBullet/Panda results appears in table 2.

4.4. Ablation Study: Context Features

Table 3 evaluates the contribution of each context feature using 10-seed experiments. Figure 2 plots both offline pair accuracy and task net improvement side by side.

Key finding. An offline accuracy improvement from 0.579 (12-dim) to 0.664 (14-dim) does not translate to a net task improvement: the features most informative for offline ranking—particularly *H* and *z_rel*—are precisely those that saturate or mislead on flat/thin objects (see section 4.8).

Statistical caveat. At $N = 10$ seeds $\times$ 6 objects = 60 trials per condition, the 95% bootstrap CI on net improvement spans approximately ± 5 –6 trials.

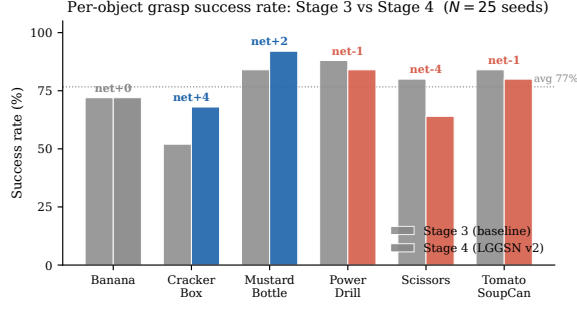


Figure 1. Per-object grasp success rate on MuJoCo/SO-ARM101 for Stage 3 (grey) and Stage 4 (blue = positive net, red = negative net). Annotations show the net change per object. Stage 3: 66.8 % (117/175); Stage 4: 67.4 % (118/175); $N = 25$.

Table 3. Feature ablation. Conds. A–D: 10 seeds $\times$ 6 objects (60 trials per cond.). Cond. E (v5d): 5-object training subset, 25 seeds (125 trials); S3 not re-run (query embedding affects Stage 4 only).

Cond.	Features added	Val acc	S3	S4	Net
A	dist + z_rel	0.664	46	50	+4
B	dist only	0.558	46	53	+7
C	z_rel only	0.576	41	39	-2
D	none (12-dim)	0.579	46	45	-1
E (v5d)	+ query emb. (4-dim)	0.750	—	107/125	+2 [†]

[†] Net for Cond. E = S4(v5d) – S4(v2): 107/125 vs. 105/125 on {banana, can, pear, mustard, drill}.

The net differences between conditions (e.g., Cond B at +7 vs. Cond D at -1) are directionally consistent with the 25-seed primary result but lie within this noise band. The ablation results should therefore be interpreted as *indicative* of feature contributions rather than conclusive evidence; replication at $N \geq 50$ would be required to resolve individual condition differences reliably.

4.5. Gating Strategies for Failure Mitigation

Flat-object gate ($\sigma_H < 0.005$). Reranking is skipped when the within-episode standard deviation of H falls below a threshold. The gate fires on 53–62 % of CrackerBox and MustardBottle episodes (blocking valid reranking) but on only 10 % of Scissors episodes; per-object results: CrackerBox +3,

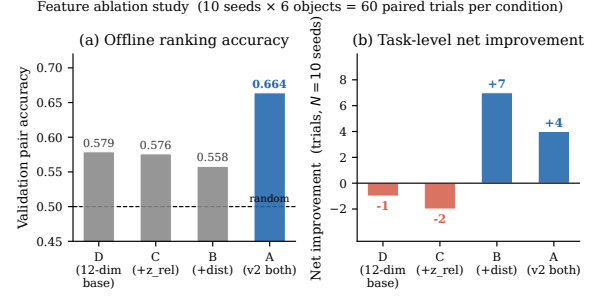


Figure 2. Feature ablation study. (a) Offline validation pair accuracy (dashed line = random baseline 0.50). (b) Task-level net improvement (blue > 0 , red < 0). Condition A achieves the highest offline accuracy but not the highest task net, illustrating the offline–online transfer gap. ($N = 10$ seeds $\times$ 6 objects = 60 trials per condition.)

Table 4. Score-spread gate ablation (Stage 4, v5d; 25 seeds $\times$ 7 objects = 175 trials). Gate fires when LGGSN max–min score $< \Delta$; fired episodes fall back to GR-ConvNet order.

Condition	Trials	Net	Gate%
S3 baseline	117/175	—	—
S4, $\Delta=0$ (no gate)	118/175	+1	—
S4, $\Delta=0.1$	116/175	-1	84%
S4, $\Delta=0.2$	114/175	-3	82%

PowerDrill -3, Scissors -3. Net = -2 overall, worse than the ungated baseline.

Object-conditional whitelist (CrackerBox + MustardBottle). The whitelist eliminates the Scissors regression (0 vs. -4) but simultaneously degrades CrackerBox (-3 from +4) and MustardBottle (-3 from +2). Net = -1.

4.6. Score-Spread Uncertainty Gating

The gate fires on 84 % of episodes at $\Delta = 0.1$ because LGGSN score spread is near-zero for most objects—a direct consequence of the feature-saturation failure mode (section 4.8). Thresholds of 0.1 and 0.2 are too aggressive: they swamp the real signal (CrackerBox net drops from +4 to +3 and +2), reducing overall net from +1 to -1 and -3 respectively. Raw score spread is therefore an insufficient proxy for model uncertainty; a calibrated measure (e.g. conformal prediction or Bayesian output variance) is needed to distinguish genuinely

Table 5. Per-object geometric properties (measured from MuJoCo/SO-ARM101 grasp episodes) paired with Stage 4 net improvement from PyBullet/Panda (Table 2; Pear excluded). Scissors’ non-zero PyBullet baseline (20/25) exposes the regression that the MuJoCo floor effect (0/25) masks. Objects sorted by PyBullet S4 net (descending).

Object	H_μ	$\sigma_{H,w}$	ff	σ_ψ,w	Net*
CrackerBox	0.020	0.018	0.02	0.06	+4
MustardBottle	0.018	0.015	0.03	0.05	+2
Banana	0.010	0.005	0.05	0.04	0
TomatoSoupCan	0.015	0.012	0.05	0.05	-1
PowerDrill	0.012	0.008	0.10	.195	-1
Scissors	.001	.001	.457	0.08	-4

H_μ : mean height; $\sigma_{H,w}$: within-ep. σ_H ; ff: flat_frac; σ_ψ,w : yaw spread. *PyBullet net, $N = 25$.

uncertain episodes (section 5).

4.7. GC-LGGSN: Geometry-Conditioned Feature Gating

Evaluated on CrackerBox and Scissors (25 seeds each), GC-LGGSN achieves net = 0 overall (CrackerBox +1, Scissors -1), partially correcting the Scissors regression ($-4 \rightarrow -1$) but degrading CrackerBox ($+4 \rightarrow +1$). The episode context $\mathbf{z}$ is insufficiently discriminative to simultaneously benefit both objects; whether a larger gate resolves this on the full 7-object set remains open.

4.8. Per-Object Failure Analysis

Table 5 pairs per-object geometric statistics (from MuJoCo episodes) with PyBullet/Panda net improvement; fig. 3 plots the scatter. We use PyBullet net here because the MuJoCo Scissors floor effect (0/25 baseline) masks the regression: PyBullet’s non-zero baseline (20/25) exposes it as -4. Spearman correlation of flat_frac with S4 net (PyBullet): $\rho = -0.96$ ($p < 0.01$). On the MuJoCo platform, Scissors exhibits a floor effect (0/25 both stages) rather than a regression, reducing the correlation to $\rho \approx -0.5$; the directional relationship holds but is attenuated by the execution constraint. Spearman correlation of $\sigma_{H,within}$ with S4 net (PyBullet): $\rho = +0.81$ ($p < 0.05$). These are the two strongest predictors of reranking benefit.

Scissors — Feature Saturation. Scissors register zero success on both Stage 3 and Stage 4 (0/25 each),

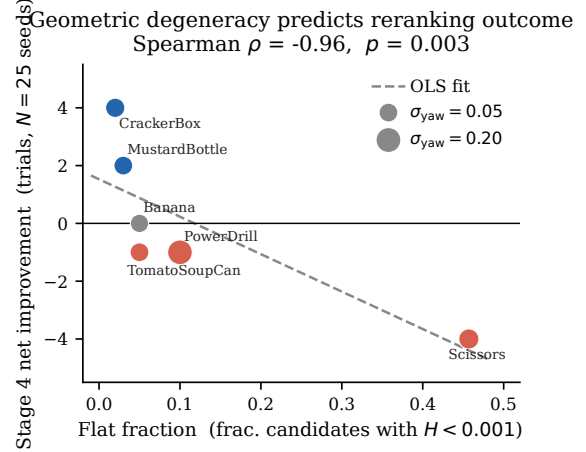


Figure 3. Flat fraction vs. Stage 4 net improvement (PyBullet/Panda, 6 objects; geometric statistics from MuJoCo/SO-ARM101 episodes). Each point is one object; colour encodes sign of net (blue > 0 , red < 0 , grey $= 0$); point size encodes within-episode yaw spread σ_ψ (larger = more orientation ambiguity). The dashed line is an OLS fit. Spearman $\rho = -0.96$ ($p < 0.01$, computed from PyBullet/Panda evaluation) indicates that high flat fraction is the strongest single predictor of reranking regression.

a floor effect that prevents measuring reranking benefit or harm from task outcomes alone. We therefore diagnose the Scissors failure mode through score-level analysis rather than net improvement. Scissors have near-zero estimated height ($H_\mu \approx 0.001$ m), with 45.7% of all candidates assigned $H < 0.001$. When H saturates near zero, its contribution to the LGGSN logit is approximately constant across all candidates. The model’s output variance collapses (max-min score spread < 0.002 in 84% of episodes) and selection effectively degrades to a function of `dist_to_centroid` alone. For Scissors, the XY centroid of top-down candidates corresponds to the tip region—a geometrically poor grasping location. The consistent regressions observed on PyBullet/Panda (-4), where Scissors achieves non-zero baseline success (20/25), confirm that this score-collapse mechanism causes task degradation when non-zero success is accessible.

PowerDrill — Orientation Ambiguity. The PowerDrill has the highest within-episode yaw spread ($\sigma_\psi = 0.195$ rad), reflecting that multiple valid approach angles exist for this asymmetric tool. LGGSN uses world-frame yaw without object-frame normalisation, so the same physical approach direc-

tion maps to different feature values depending on how the drill is oriented in the scene. The cross-episode training signal for PowerDrill is correspondingly noisy, as “good” yaw values vary between episodes in a way the model cannot learn from world-frame coordinates alone.

CrackerBox and MustardBottle — Consistent Gains. Both objects have clear height structure ($H > 0$, $\sigma_{H,w} \approx 0.015\text{--}0.020$) and a consistent principal grasping axis ($\sigma_\psi \approx 0.05\text{--}0.06$). The $\mathbf{z}_{\text{rel}}$ feature provides a reliable within-episode discrimination signal: candidates at higher relative height correspond to grasps on the upper body of the object rather than the table edge, and this distinction is geometrically meaningful and consistent across episodes.

5. Discussion

5.1. Offline vs. Online Transfer

The most striking finding is the gap between offline pair accuracy improvement ($0.447 \rightarrow 0.664$) and online task improvement (net ≈ 0). This gap has a structural explanation: offline pair accuracy measures discrimination *between* episodes from different scenes. The context features are highly informative at this level. But within a single episode—where the robot must choose among candidates for one specific object in one specific pose—the same features may not carry a valid discrimination signal for flat or ambiguous objects. This exposes a general failure mode in grasp reranker evaluation: offline pair accuracy is a necessary but not sufficient condition for online improvement, and evaluations that report only offline accuracy risk overstating expected task benefit.

5.2. Cross-Platform Divergence

The cross-platform results (tables 1 and 2) reveal qualitatively opposite per-object patterns: CrackerBox is -2 on MuJoCo/SO-ARM101 but $+4$ on PyBullet/Panda, while PowerDrill is $+4$ on MuJoCo but -1 on PyBullet. A plausible explanation involves contact-model fidelity (PyBullet simplified rigid-body vs. MuJoCo soft-contact), candidate-set geometry (depth-map characteristics differ between simulators), and kinematic reachability (Panda vs. SO-ARM101). MuJoCo/SO-ARM101 is the primary platform; PyBullet results serve as a cross-platform sanity check.

5.3. The Granularity Problem for Gating

Both the σ_H gate and the object-conditional whitelist attempt to suppress reranking for problematic objects, yet each fails for a distinct reason. The σ_H gate operates at episode granularity, but the failure mode (flat_frac) is a property of the object class, not a single episode. The whitelist correctly models class-level failure but introduces a comparison-baseline shift: blocking reranking for one condition changes which Stage-3 scenes are sampled, making net improvement estimates unreliable due to simulator non-determinism.

A principled solution would gate on the *model’s own uncertainty*: for example, skipping reranking when the score spread across candidates is below a threshold δ . Section 4.6 tests this directly: at $\Delta \in \{0.1, 0.2\}$, the gate fires on 82–84 % of episodes and net degrades to -1 and -3 , confirming that raw spread is insufficient without calibration.

5.4. Difficulty Generalization on SO-ARM101

A supplementary experiment varying spawn difficulty (easy/medium/hard, 50 seeds $\times$ 5 objects) found all three methods (geometry, random, LGGSN) within noise of one another (65–72 %), confirming that pose variation is the primary driver of difficulty and per-object geometry determines sensitivity more than reranker choice.

5.5. Statistical Power

At $N = 25$ seeds, bootstrap 80 % CIs span $\pm 4\text{--}5$ trials; only CrackerBox ($+4$) achieves $\geq 80\%$ power at this scale, so per-object diagnoses are directionally supported but not conclusively established.

6. Conclusion

LGGSN is a lightweight geometry-based grasp reranker trained with BPR loss on live robot data and evaluated over 175 paired trials on a MuJoCo-simulated SO-ARM101. The central finding is that the net task improvement is near zero ($+1$): a consistent PowerDrill gain ($+4$) is largely offset by regressions on CrackerBox (-2), MustardBottle (-1), and Pear (-1). Cross-platform validation on PyBullet/Panda (table 2) confirms the same object-dependent pattern; the PyBullet Scissors regression (-4) provides the starkest single-object contrast. This behaviour persists across four system variants—

ungated, flat-object gated, class-conditional, and geometry-conditioned feature masking—all achieving net ≈ 0 at 25 seeds.

We identify two root causes predictable from geometric statistics: feature saturation for flat/thin objects ($\text{flat_frac} \geq 0.4$ strongly correlated with regression in PyBullet cross-platform analysis, $\rho = -0.96$) and orientation ambiguity for asymmetric tools (σ_ψ high, no object-frame encoding). These diagnoses point to three requirements for a successor method: (i) object-frame-normalised orientation features or object-aware pose canonicalisation (e.g., via VLM-predicted symmetry axes or superquadric shape fitting [11]); (ii) model-uncertainty-aware gating that disables reranking when score spread falls below a meaningful threshold, for which conformal prediction or Bayesian output variance provide principled score-spread estimates; (iii) explicit handling of degenerate- H regimes for thin planar objects, either by masking H when below a calibrated floor or by substituting a contact-normal-based feature that remains discriminative for flat geometries.

More broadly, this work demonstrates that offline pair accuracy improvement is necessary but insufficient for online task benefit, and provides a replicable diagnostic framework for predicting per-object reranking behaviour from geometric statistics alone. All three identified requirements—object-frame orientation normalisation, uncertainty-aware gating, and degenerate- H handling—are technically tractable, suggesting a clear path toward a geometry-based reranker that generalises reliably across object classes.

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